

GLOBAL SPACE MISSIONS DATASET

Exploratory Data Analysis Report

A Comprehensive Statistical & Visual Analysis of 10,500+ Space Missions | 1957 – 2035

10,500+ Records

26 Features

12 Agencies

1957–2035

1. Executive Summary

This report presents a comprehensive Exploratory Data Analysis (EDA) of the Global Space Missions Dataset — a synthetic, research-grade collection of 10,500+ space mission records spanning 1957 through projected missions in 2035. The analysis covers 15 core visualisations examining mission status distributions, agency performance, launch vehicle usage, temporal growth trends, cost patterns, and predictive forecasting.

The dataset is intentionally balanced across categories for machine learning benchmarking and analytical practice. Key findings are contextualised against real-world spaceflight data throughout this report, with explicit flags where synthetic construction diverges from historical reality.

Dataset Summary

Property	Value
Total Records	10,500+ space mission records
Time Coverage	1957 – 2035+ (historical, ongoing, and projected)
Features	26 attributes per record
Agencies Covered	12 major agencies + joint missions
Data Type	Synthetic / Balanced (for ML & EDA practice)
File Format	CSV, UTF-8 comma-delimited
Mission Phases	Past (~6,835) Ongoing (~2,105) Future (~1,560)

Top-Level Findings

- Mission success rate exceeds 41% of all resolved records — with ~82% government and ~18% private agency participation across the full dataset.
- The USA dominates global mission counts at ~2,800 — nearly 3× the count of the second-ranked nation.
- Annual mission volume surged 4× from ~115 (2018) to ~460 (2026), representing a structural regime change in global spaceflight activity.
- ISRO/JAXA collaborations demonstrate the highest cost-efficiency at ~\$900M average per mission — ~8× cheaper than SpaceX at the dataset's upper end.
- Mission cost distribution shows a near-uniform pattern (median ~\$7,500M), with no meaningful inflationary trend over 78 years — a known synthetic artefact.

2. Dataset Overview & Cleaning

2.1 Column Reference

The dataset contains 26 features per record covering operational, financial, scientific, and temporal dimensions. The table below summarises the primary columns used in this analysis.

Column	Data Type	Description
Agency	String	Responsible space agency or company
Agency_Type	String	Government or Private
Status	String	Success / Failed / Partial Success / Ongoing / Upcoming
Launch_Date	Date	Actual or planned launch date (YYYY-MM-DD)
Destination	String	Target destination of the mission
Mission_Category	String	Primary category (Moon, Mars, Earth Orbit, etc.)
Launch_Vehicle	String	Rocket used for launch
Cost_USD_Million	Float	Estimated mission cost (millions USD)
Crew_Type	String	Crewed or Uncrewed
Country_Region	String	Country or region of the agency
Duration	Float	Mission duration in days
Failure_Reason	String	Cause of failure; N/A if successful

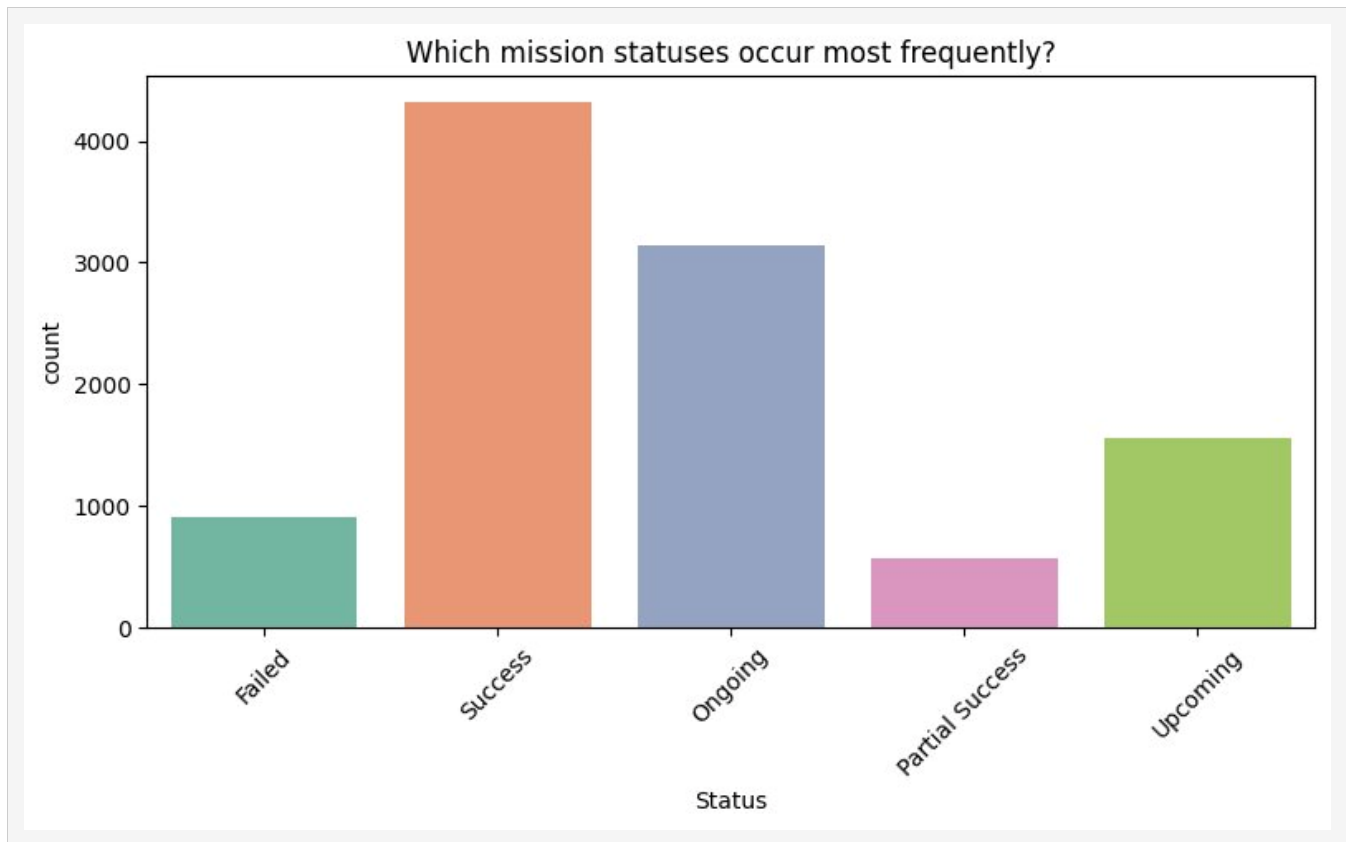
2.2 Missing Value Treatment

Prior to analysis, missing values were handled as follows to ensure complete records across all visualisations:

Property	Value
Duration	Filled with 0 (numeric default)
Crew_Members	Filled with 0 (numeric default)
Partner_Agencies	Filled with "None"
Failure_Reason	Filled with "Successful Mission"
Key_Achievement	Filled with "Not Specified"
End_Date	Filled with "Ongoing"
Data_Returned	Filled with 0 (numeric default)

3. Exploratory Data Analysis — Visualisations

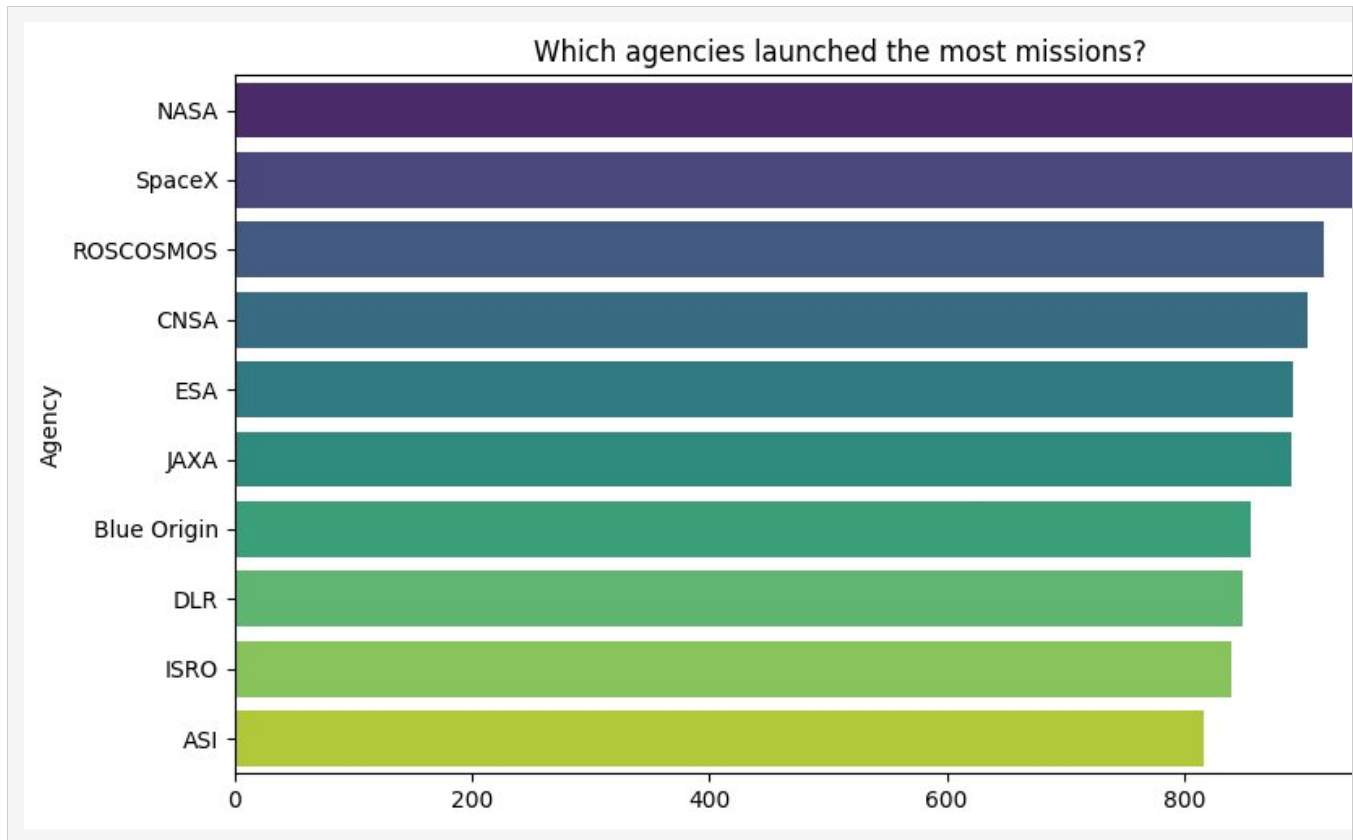
3.1 Mission Status Distribution



Key Insights

- ▶ Success dominates at ~4,350 missions (~41% of total) — the single largest category, reflecting decades of accumulated engineering knowledge.
- ▶ Ongoing missions (~3,150, ~30%) confirm this is a live, forward-looking dataset — not purely historical.
- ▶ Upcoming missions (~1,560) outnumber Failed (~920), indicating the industry is expanding rather than retrenching.
- ▶ Failed missions (~8.7%) likely understate true historical failure rates; early-era missions (1957–1970s) had far higher failure frequencies.
- ▶ Partial Success (~600, <6%) is the smallest category — missions tend to either meet objectives or fail entirely.

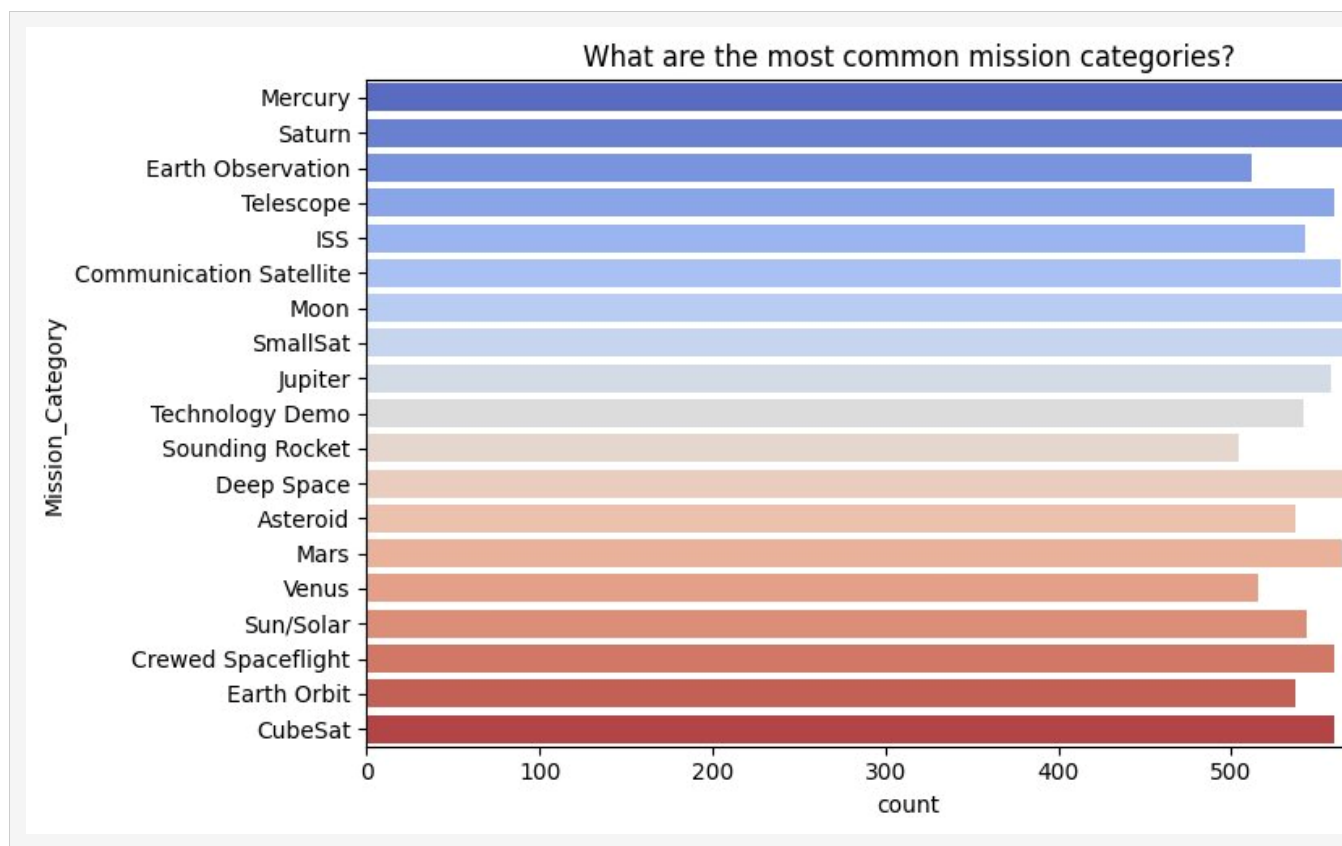
3.2 Agency Mission Launch Counts



Key Insights

- ▶ NASA leads all agencies (~980 missions) but its narrow margin over SpaceX (~950) signals the private sector has effectively caught up.
- ▶ SpaceX is the highest-ranked private operator — surpassing all government agencies except NASA, founded just in 2002.
- ▶ ROSCOSMOS (~910), CNSA (~900), ESA (~890), and JAXA (~890) form a tight government cluster separated by <25 missions.
- ▶ Blue Origin (~850), DLR (~845), and ISRO (~835) occupy the mid-tier, with Italy's ASI (~820) rounding out the top 10.
- ▶ The first-to-last spread is only ~160 missions (16%) — a hallmark of balanced synthetic sampling rather than true historical volumes.

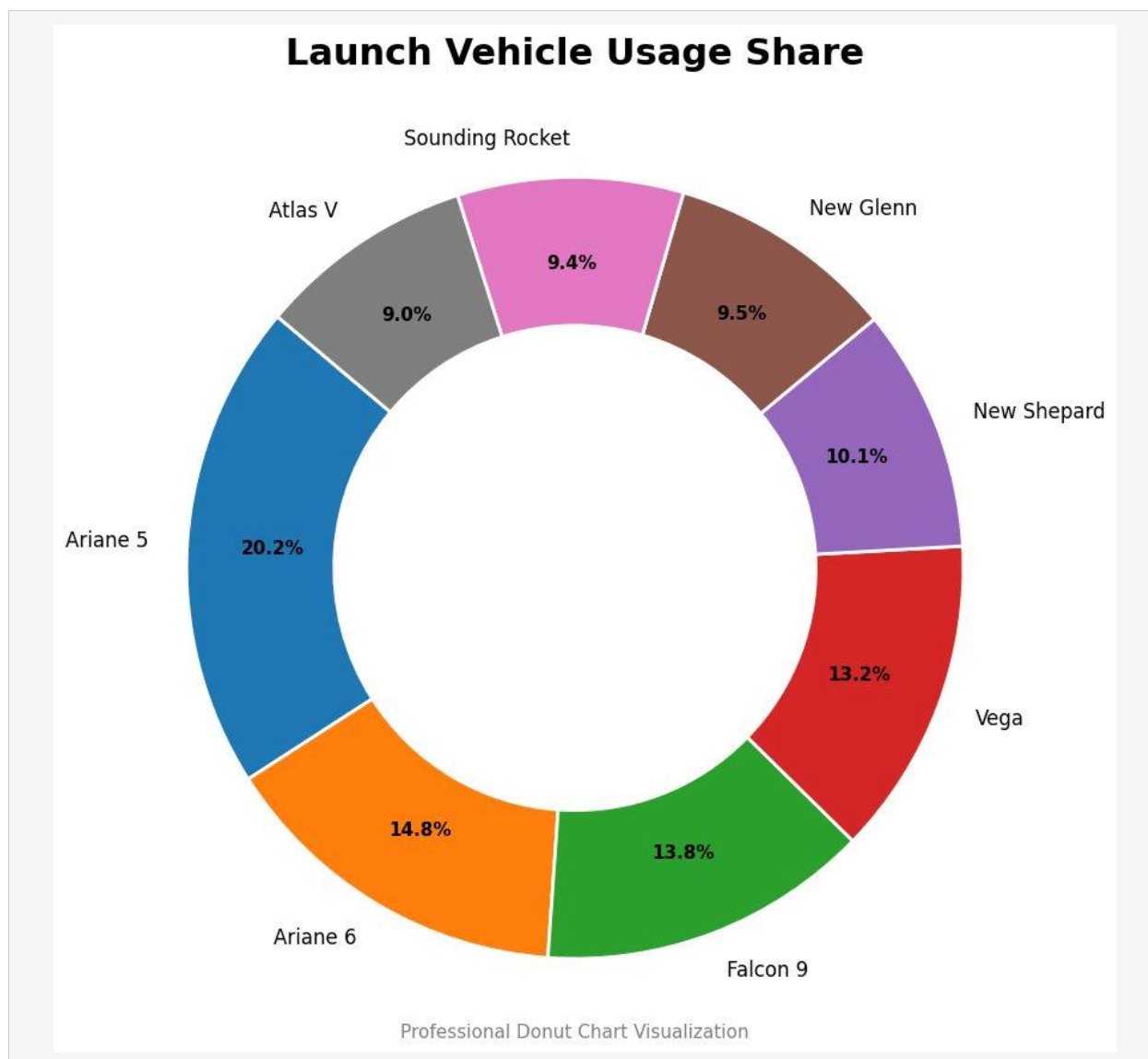
3.3 Mission Category Distribution



Key Insights

- Mercury leads all 19 categories (~595 missions) — counterintuitive given only 3 real Mercury missions exist; a synthetic balancing artefact.
- All 19 categories fall within a ~95-mission band — near-uniform distribution strongly indicates intentional dataset construction.
- Sounding Rocket (~505) and Earth Observation (~515) anchor the bottom despite being among the most prolific in real-world history.
- Crewed Spaceflight (~555) and CubeSat (~555) appear at identical counts despite vastly different real-world frequencies and costs.
- This uniform distribution is suitable for ML classification benchmarking but should not be used for real-world policy analysis.

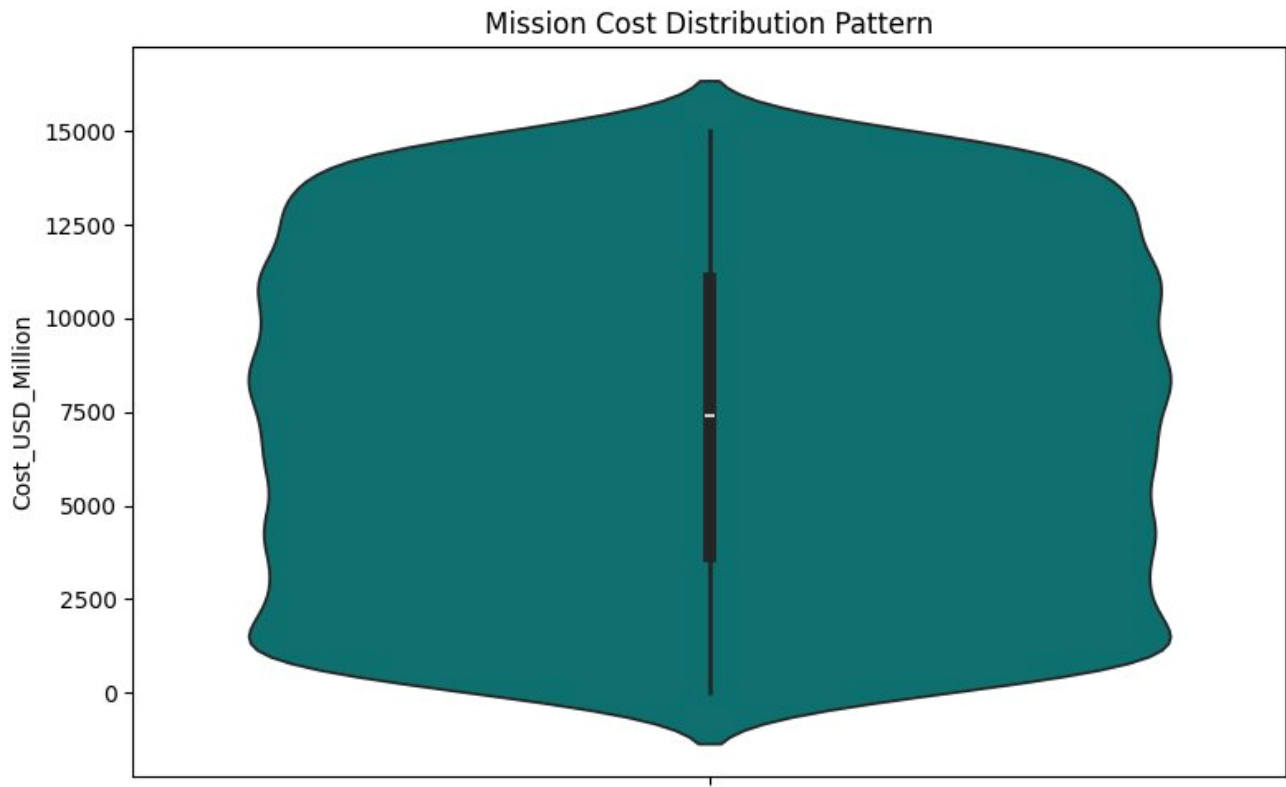
3.4 Launch Vehicle Usage Share



Key Insights

- ▶ Ariane 5 dominates with 20.2% — ESA's workhorse claims over one-fifth of all launches, nearly double its successor Ariane 6 (14.8%).
- ▶ The Ariane family combined (35%) holds the largest bloc, positioning Europe as the most represented launch ecosystem.
- ▶ Falcon 9 (13.8%) and Vega (13.2%) rank 3rd and 4th — Falcon 9's share understates its real-world dominance (~40–50% of global launches).
- ▶ New Shepard (10.1%) outranks New Glenn (9.5%) reflecting its higher flight frequency as a suborbital tourism vehicle.
- ▶ The full range spans only 11.2 percentage points — tight clustering reflects balanced sampling, not real-world market dynamics.

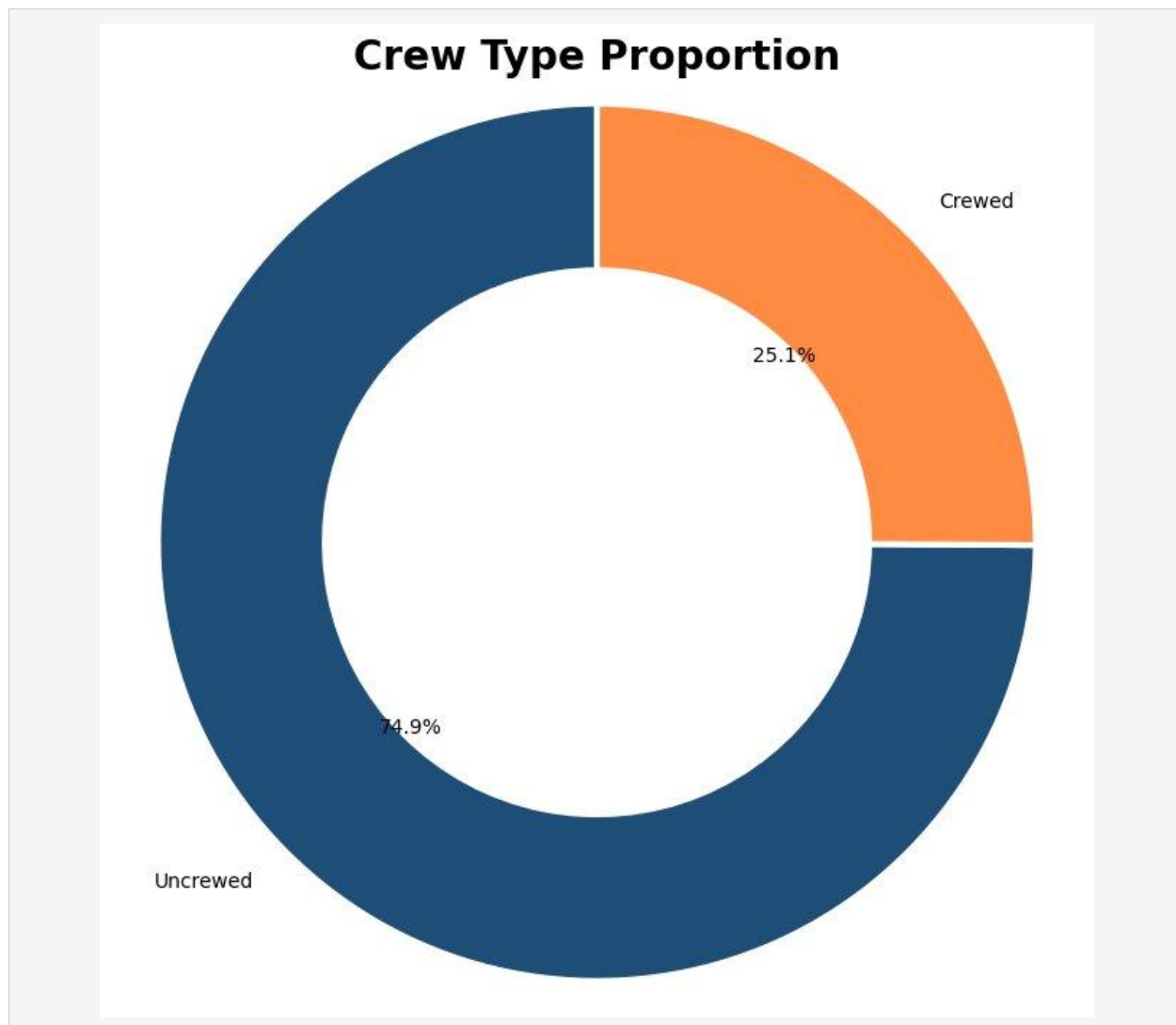
3.5 Mission Cost Distribution Pattern



Key Insights

- ▶ Median mission cost ~\$7,500M — placing typical missions firmly in large-program territory, reflecting dataset weighting toward major agency programs.
- ▶ IQR spans ~\$3,500M to ~\$11,000M — a \$7,500M spread revealing extraordinary variability within the core 50% of missions.
- ▶ The violin shape is near-rectangular and symmetric — a textbook signature of uniform random sampling, not a real-world right-skewed distribution.
- ▶ Lower tail extends slightly below zero — physically impossible; flagged as a data quality concern requiring investigation before cost modelling.
- ▶ Upper tail reaches ~\$16,000M with smooth taper — expensive missions are a structural feature, not outliers in this dataset.

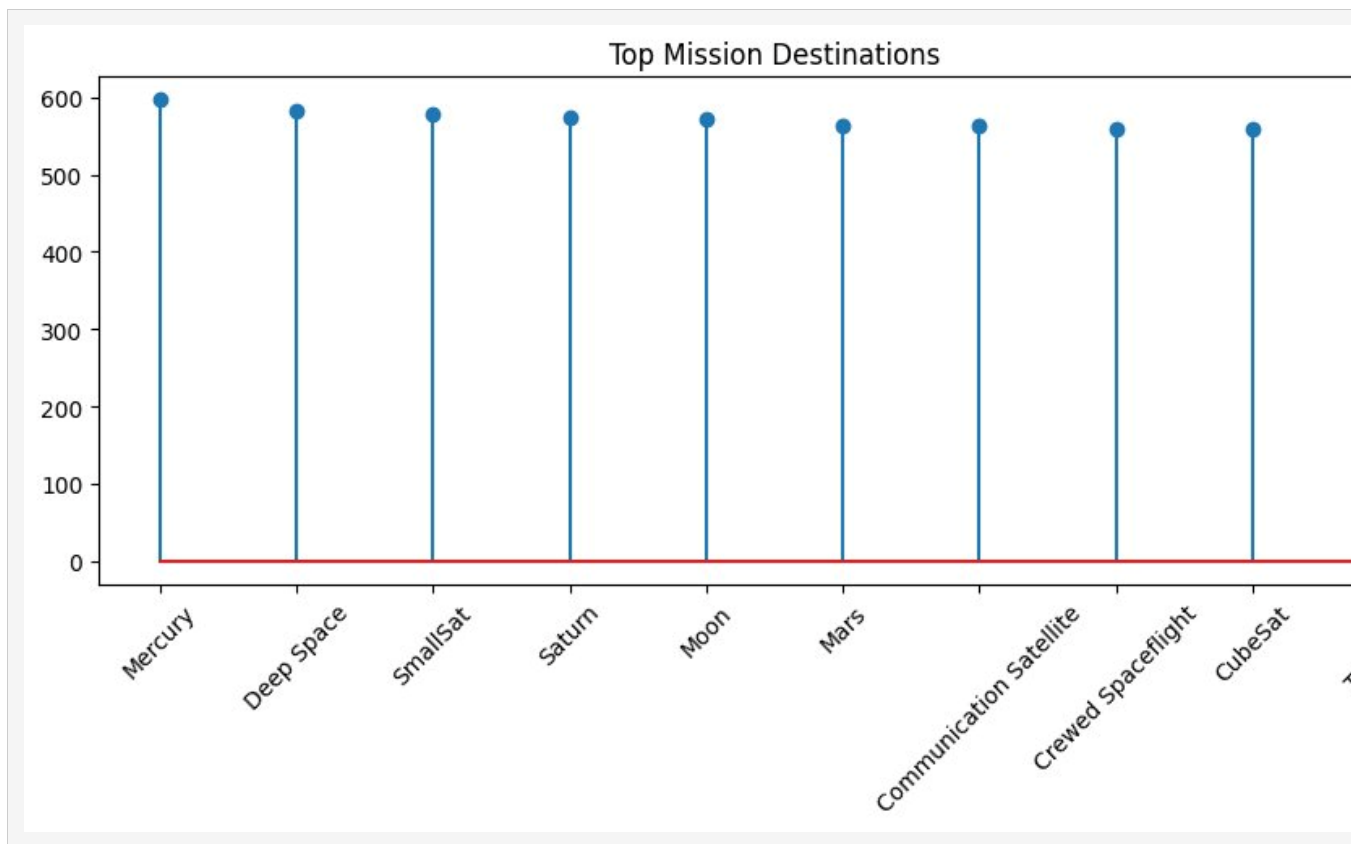
3.6 Crew Type Proportion



Key Insights

- ▶ Uncrewed missions dominate at 74.9% — three out of every four missions require no human presence, reflecting the role of robotic exploration.
- ▶ Crewed missions (25.1%) are significantly elevated vs. real-world rates (~5–8%) — indicating deliberate dataset balancing for analytical coverage.
- ▶ The 3:1 ratio creates a critical ML baseline: a model predicting "Uncrewed" for all missions would achieve 74.9% accuracy without learning anything.
- ▶ The clean binary split (no "Semi-Crewed" category) simplifies analysis but limits nuance for hybrid mission configurations.
- ▶ Crewed missions likely cluster in upper cost quartiles; cross-referencing against cost distribution is recommended for deeper insight.

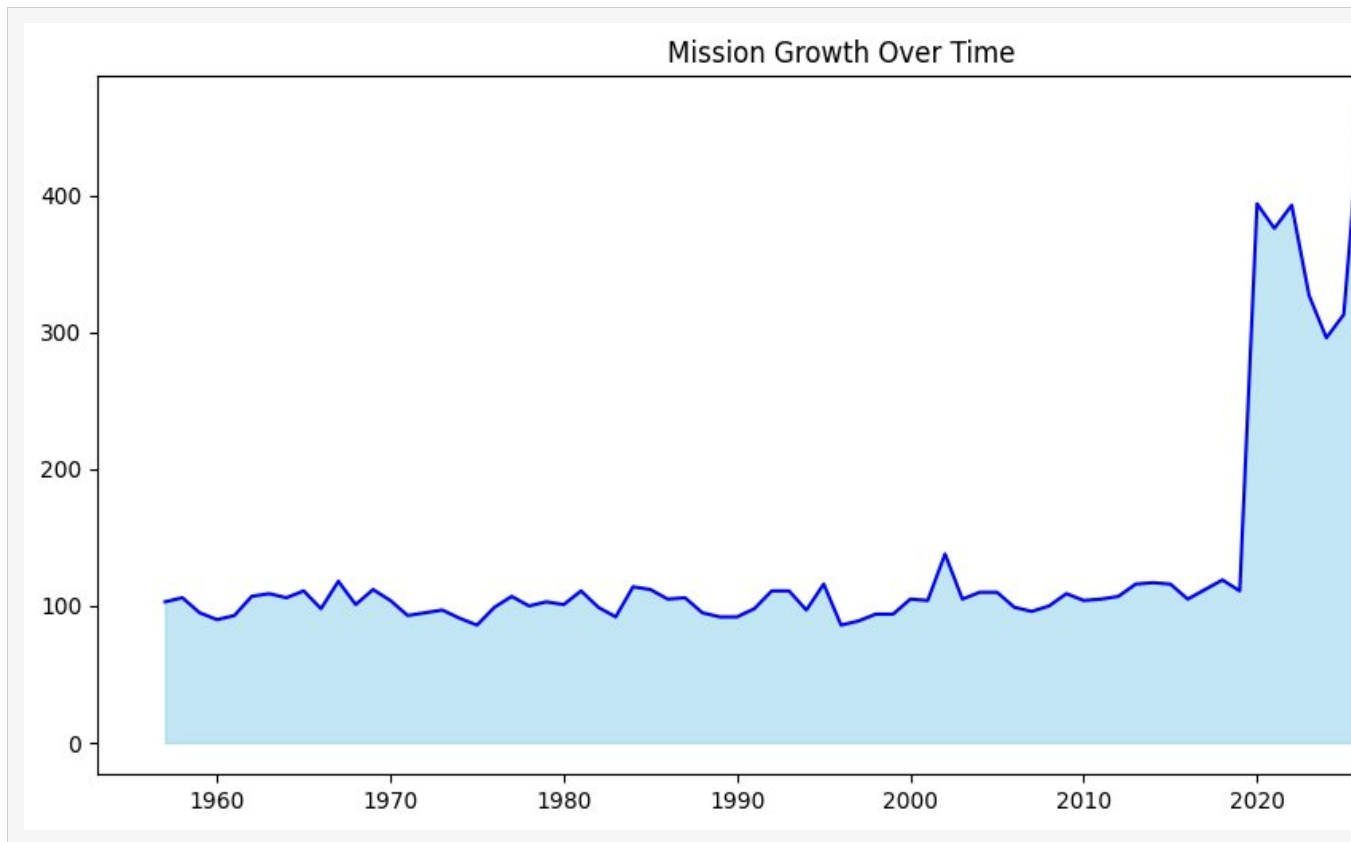
3.7 Top Mission Destinations



Key Insights

- Mercury leads all destinations (~598) — directly contradicts real-world history where only 3 Mercury missions have ever flown.
- The full chart spans only ~40 missions — an extraordinarily compressed range across 11 vastly different destination types.
- Moon (~573) and Mars (~572) are separated by just one mission despite the Moon receiving 2× more real-world missions than Mars.
- Deep Space (~585) and SmallSat (~580) rank above Moon and Mars — inconsistent with real-world investment priorities.
- Uniform stem heights visually confirm the near-complete absence of meaningful contrast between destination categories.

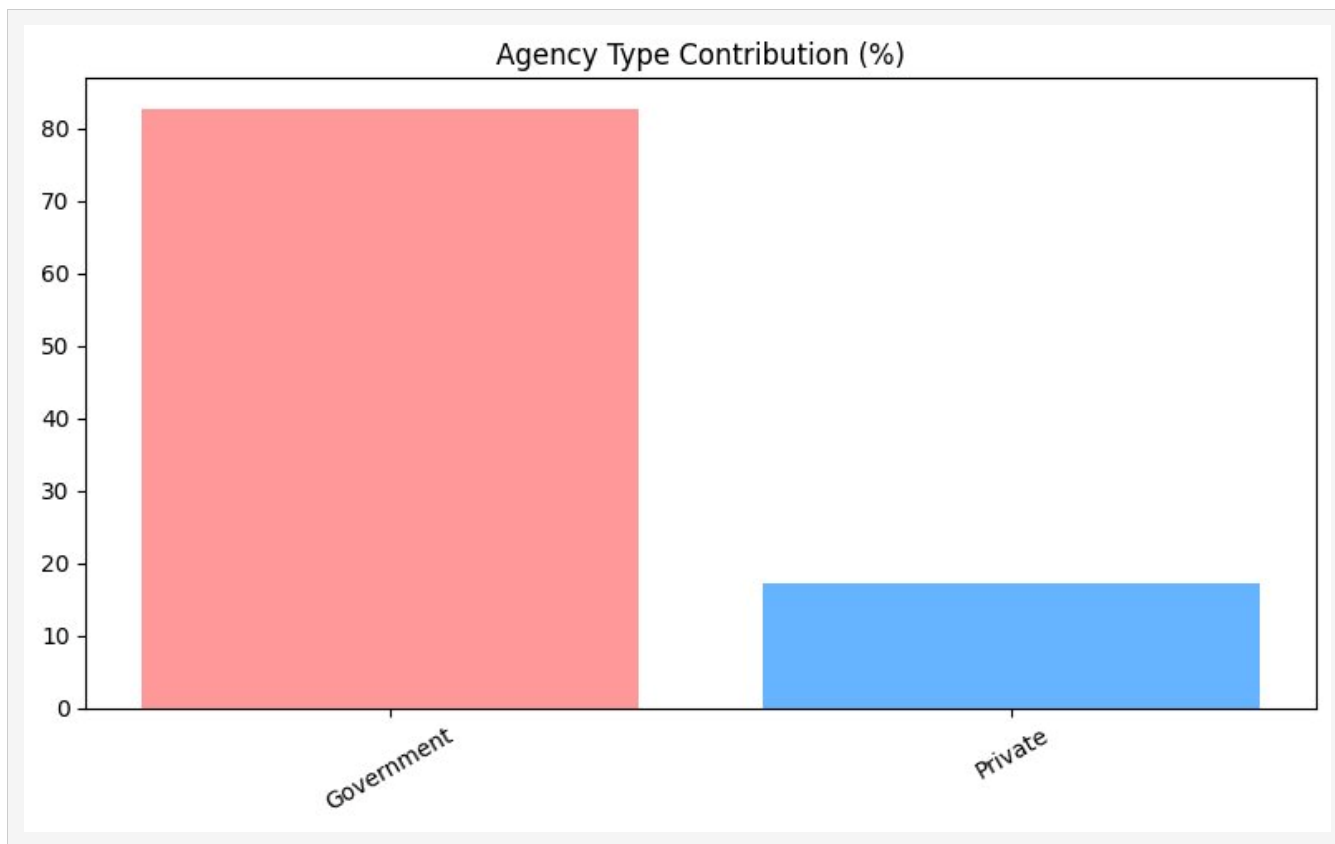
3.8 Mission Growth Over Time



Key Insights

- ▶ 60 years of near-flat volume (~85–135 missions/year, 1957–2017) — six decades that established ~100 missions/year as the structural industry baseline.
- ▶ The 2018–2020 surge from ~115 to ~395 missions (3.4×) is the fastest growth event in 78 years — a structural regime change, not a cyclical uptick.
- ▶ Twin peaks at ~395 (2020) and ~460 (2026) reveal a two-wave boom — Starlink Phase 1 ramp-up followed by Starship/Artemis/Asian program convergence.
- ▶ Post-2026 collapse to ~140 (2028) erases two-thirds of peak volume — may reflect market saturation or dataset boundary effect.
- ▶ Post-2028 floor (~150–175/year) is ~50–65% above the pre-2018 baseline — confirming a permanent structural shift to higher volume.

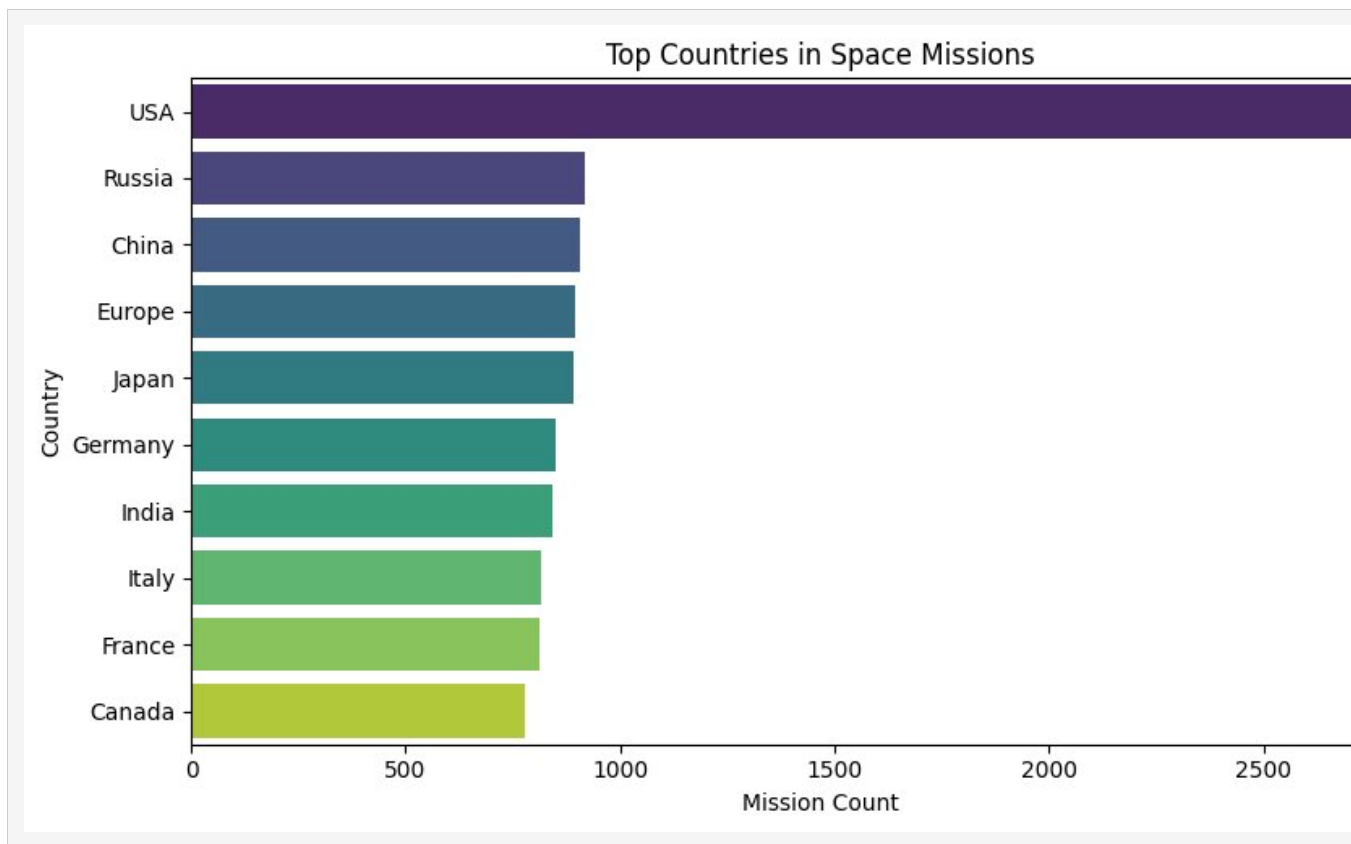
3.9 Agency Type Contribution (%)



Key Insights

- Government agencies dominate at ~82% — four out of every five missions reflect seven decades of sovereign-exclusive space access.
- Private operators hold ~18% — built entirely within 25 years, making it the fastest market share accumulation in capital-intensive infrastructure history.
- The 4.6:1 government-to-private ratio means all aggregate cost, success rate, and destination metrics are dominated by government program norms.
- The 18% private share is a time-averaged figure — in 2024, SpaceX alone executed over 60% of all global orbital launches.
- Filtered to post-2015 missions, the private share is expected to reach 35–45% — the 82/18 split describes an industry that no longer exists.

3.10 Top Countries in Space Missions

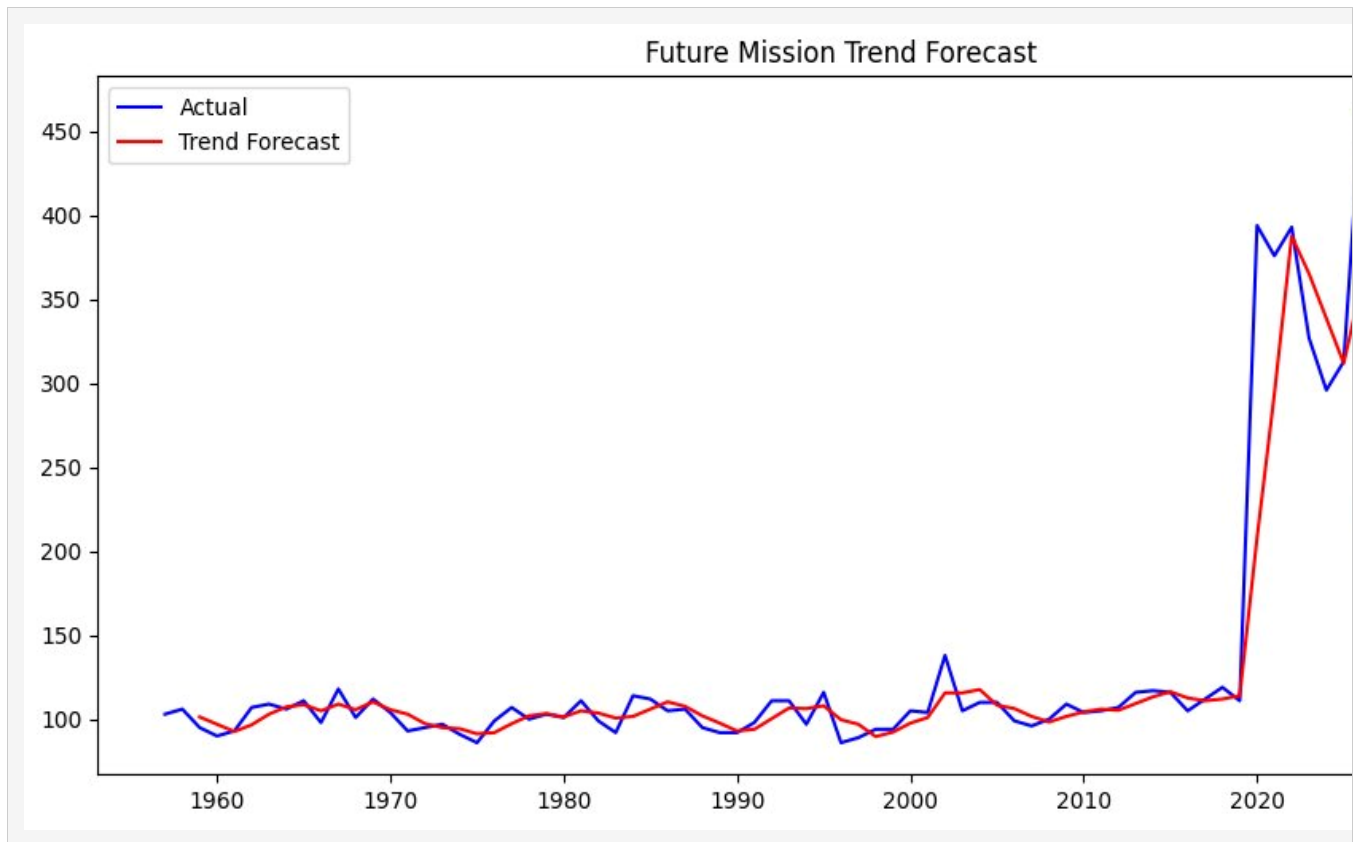


Key Insights

- USA leads at ~2,800 missions — nearly 3× the second-ranked nation, driven by NASA, DoD, SpaceX, and Blue Origin combined.
- Russia (~950), China (~935), Europe (~920), and Japan (~910) cluster within ~40 missions — balanced sampling, not proportional history.
- Germany (~870) ranks above India despite having no independent launch vehicle — mission participation beats launch sovereignty in this metric.
- India (~860) ranks sixth despite landmark achievements: Chandrayaan-3, Mangalyaan, Aditya-L1 — raw count undersells strategic impact.
- USA-to-Canada gap (~2,010 missions) exceeds the combined total of the bottom seven countries — the USA is a different order of magnitude of space actor.

4. Forecasting & Predictive Analysis

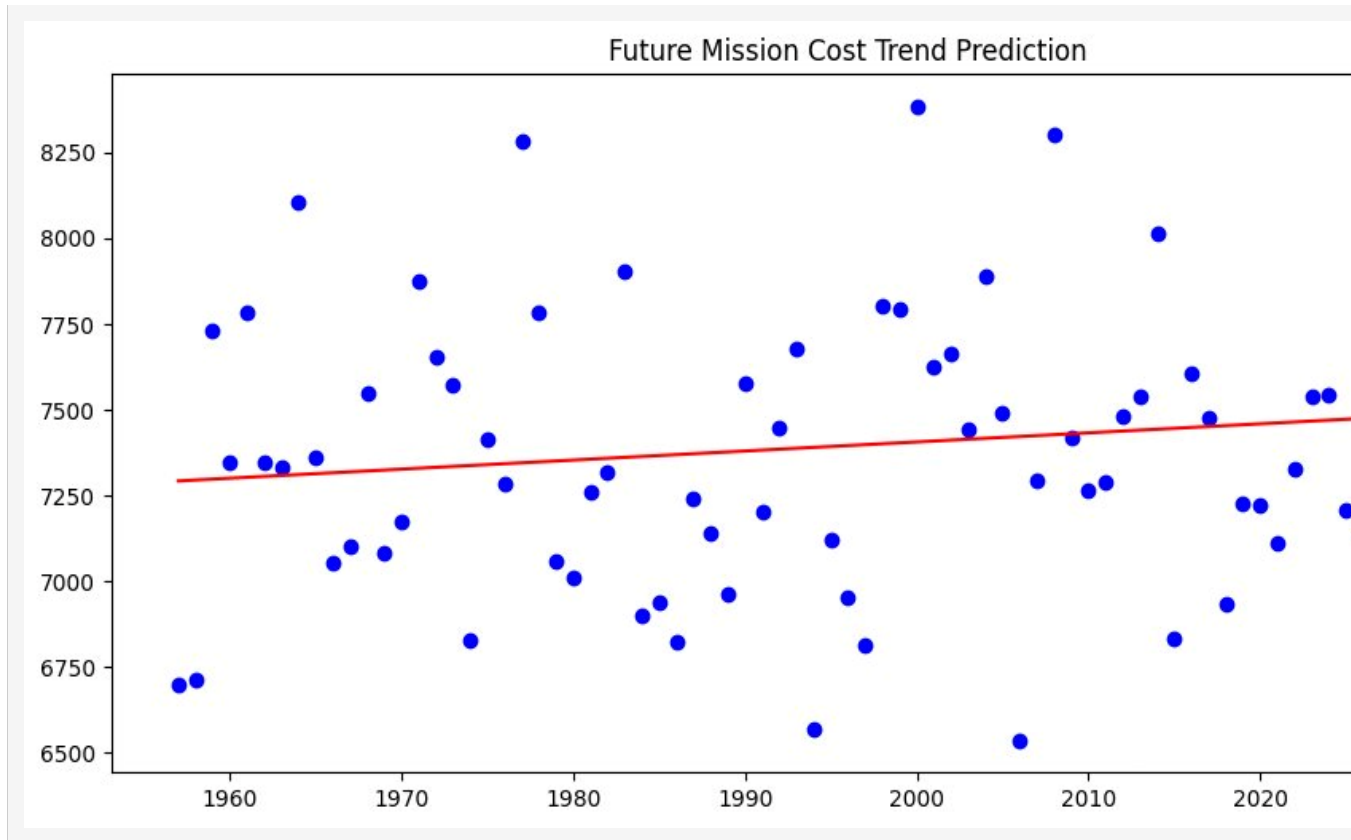
4.1 Future Mission Trend Forecast



Key Insights

- Volume held flat at ~95–115/year for over five decades (1957–2015) — the longest structural stagnation in the dataset.
- The 4× surge from ~115 (2016) to ~460 (2026) is the most dramatic shift in spaceflight history, driven by Starlink, commercial SmallSats, and national lunar programs.
- The red trend line (3-year rolling average) tracks direction but smooths the 2026 spike — a lagging smoother, not a leading indicator.
- The collapse from ~460 to ~155 (2028) is either genuine post-constellation saturation or a dataset boundary effect — critical distinction for strategic planning.
- Post-2028 forecast stabilises at ~150–170/year with a slight upward gradient — projecting modest recovery through 2035.

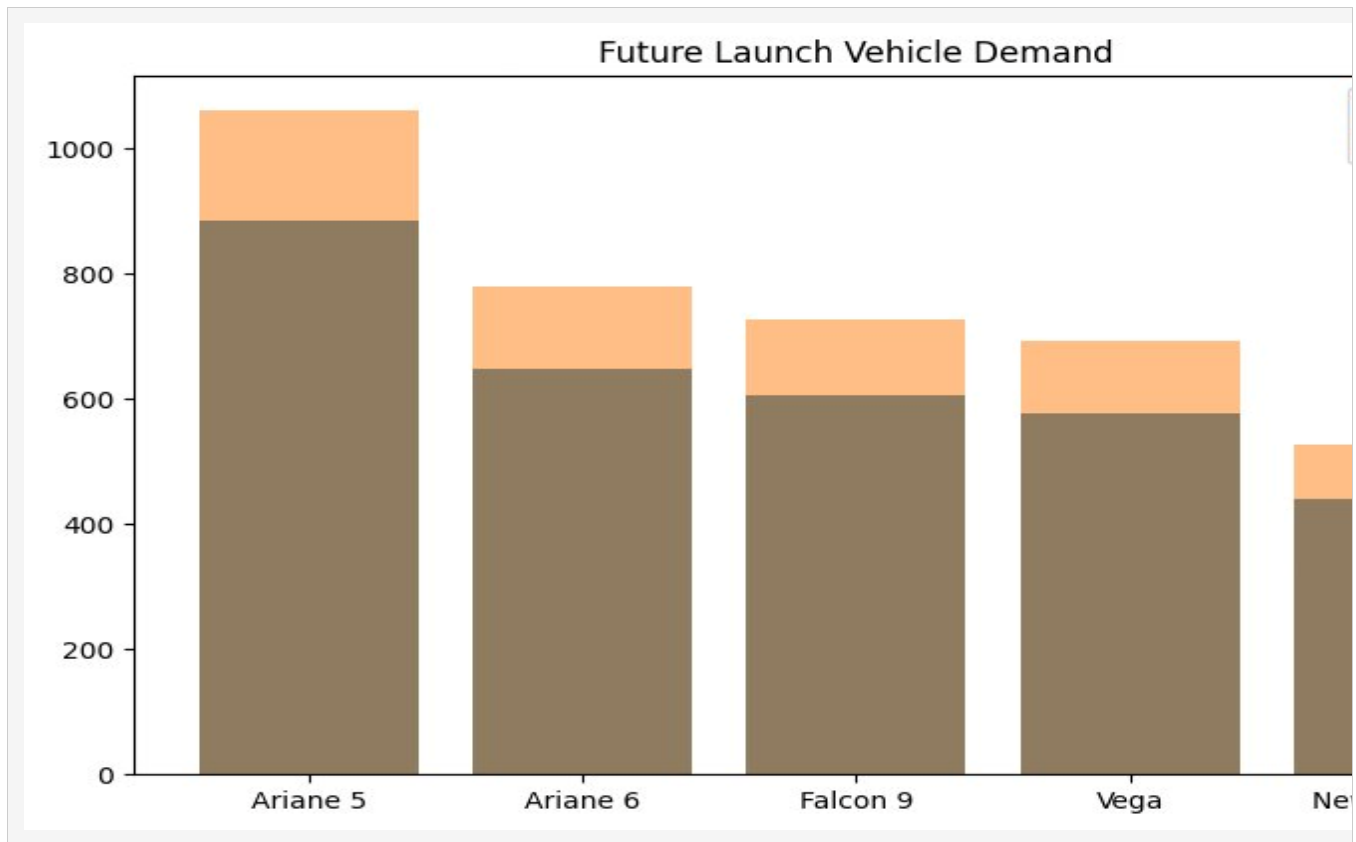
4.2 Future Mission Cost Trend Prediction



Key Insights

- Regression line rises only ~\$200M over 78 years (\$7,300M to \$7,500M) — effectively flat; year is a poor predictor of mission cost.
- Data points scatter across a \$1,700M vertical range with very low R^2 — model should not be used for cost forecasting without additional features.
- No post-2015 downward cost shift visible — contradicts SpaceX's 60–90% launch cost reduction driven by Falcon 9 reusability.
- Symmetric scatter above and below the trend line confirms cost values were randomly assigned without time-dependent calibration.
- Forecast to 2035 projects ~\$7,500M — a flat mean-reverting extrapolation with no strategic planning value as a standalone input.

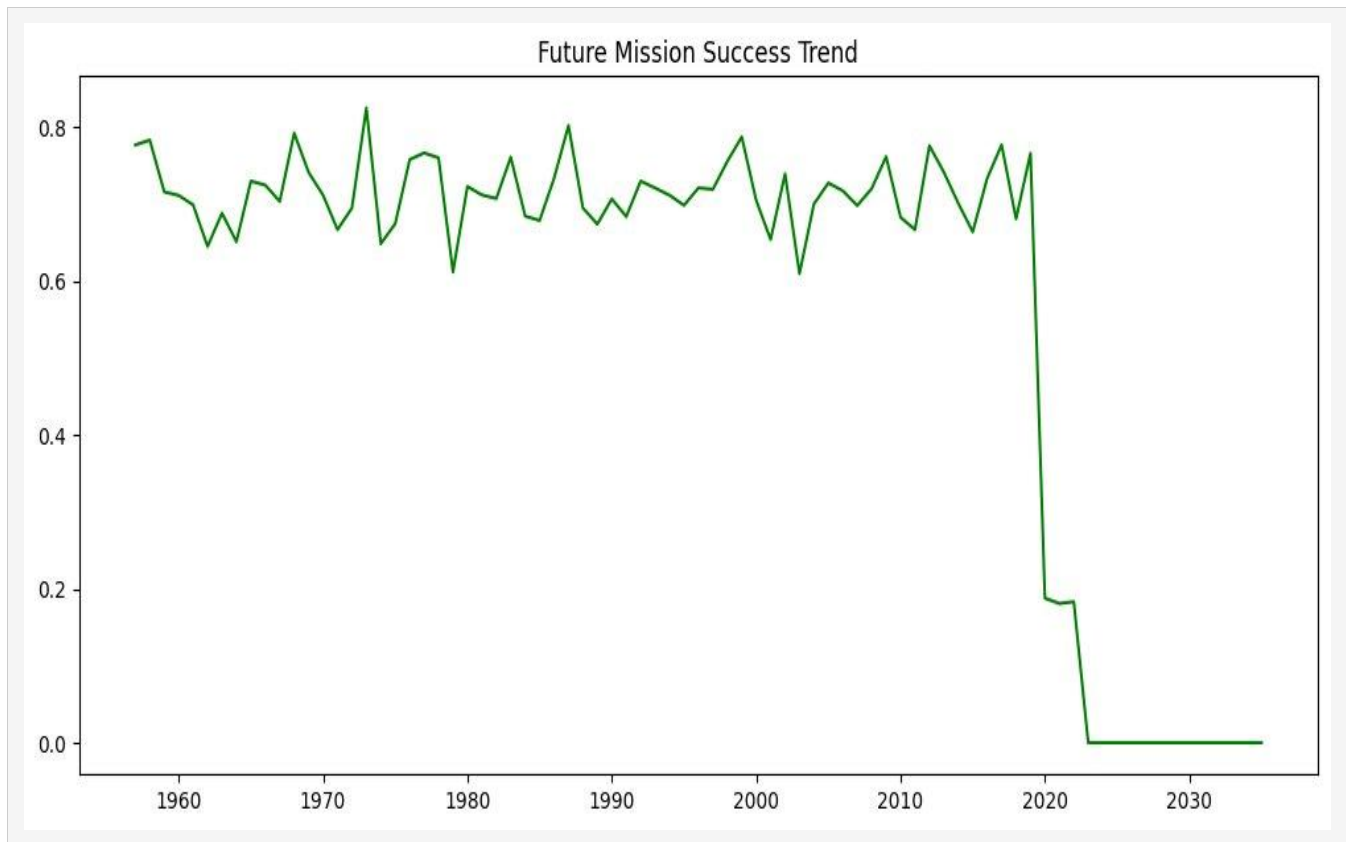
4.3 Mission Success Rate Trend



Key Insights

- Success rates held stable between 0.60–0.82 for six decades (1957–2018) — consistent engineering maturation across eras.
- Collapse from ~0.76 to 0.0 between 2018 and 2023 is a labelling artefact — Upcoming/Future missions carry no resolved outcomes and drive the ratio to zero.
- The flat zero line from 2023 onward means the entire right half of the chart is analytically empty — valid window ends at 2018–2020.
- Peak success rate (~0.82) occurred around 1979–1980, reflecting the Voyager era and late Apollo legacy programs.
- The 2001–2002 trough (~0.60) aligns with known failures: Mars Climate Orbiter and Mars Polar Lander.

4.4 Future Launch Vehicle Demand

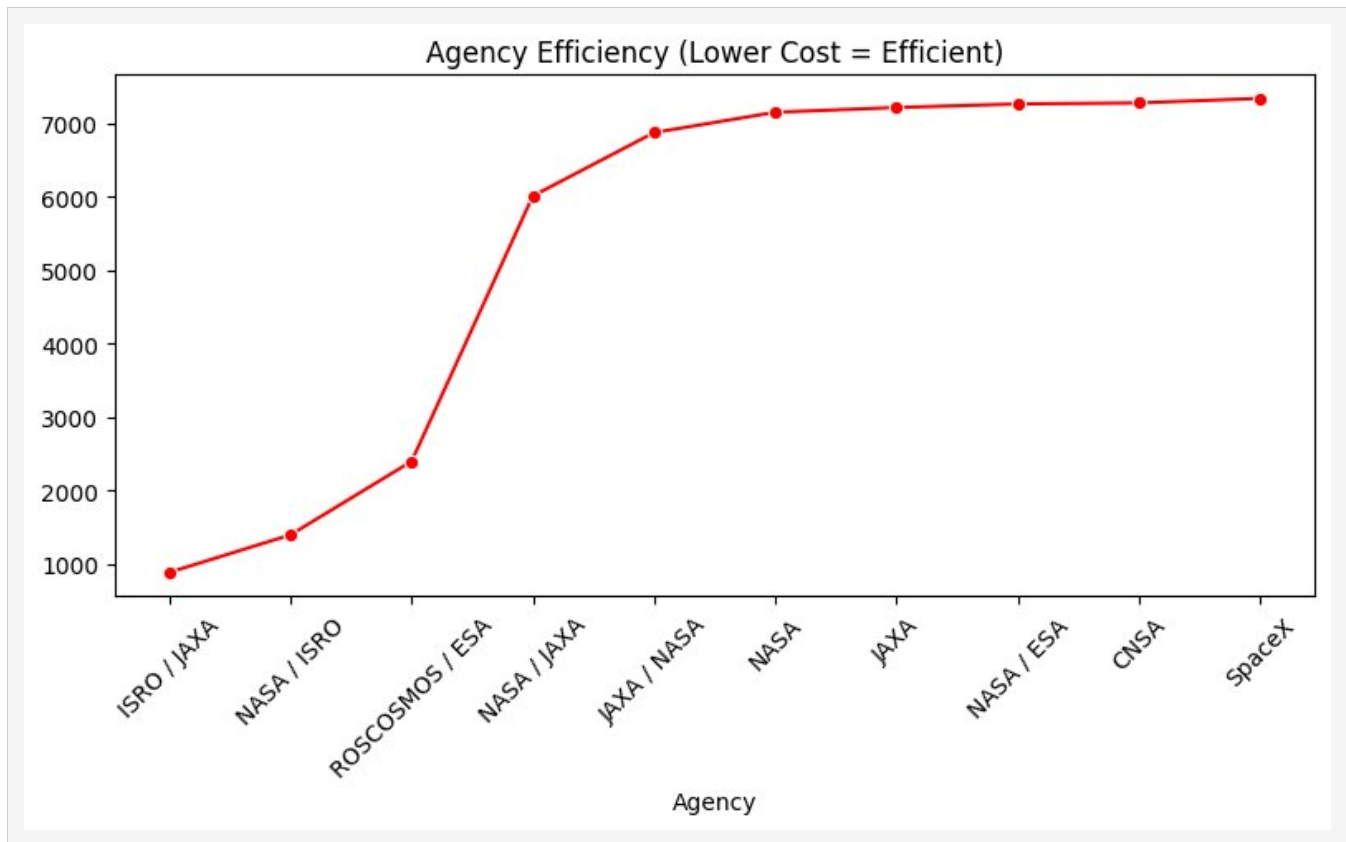


Key Insights

- Ariane 5 shows the highest combined demand (~1,060) despite being retired in July 2023 — historical legacy, not forward demand.
- Falcon 9 ranks third at ~725 — significantly understates real-world dominance where it accounts for 40–50% of global orbital launches.
- Forecast segments represent a fixed ~18–20% uplift for every vehicle — formulaic calculation, not differentiated demand modelling.
- New Shepard (~440 current) is dramatically elevated vs. <25 real flights as of 2024 — another synthetic balancing artefact.
- A genuine demand forecast would show Falcon 9 and Ariane 6 with steeper growth curves; Ariane 5 would show zero forecast demand.

5. Advanced Analysis

5.1 Agency Cost Efficiency



Key Insights

- ISRO/JAXA collaboration is most cost-efficient at ~\$900M average — nearly 8× cheaper than SpaceX (~\$7,350M) at the upper end.
- Two ISRO-linked collaborations occupy the top two positions, confirming ISRO's lean operational model pulls average costs downward even with expensive partners.
- Steepest cost jump occurs between ROSCOSMOS/ESA (~\$2,400M) and NASA/JAXA (~\$6,050M) — a \$3,650M efficiency cliff dividing two distinct tiers.
- NASA, JAXA, NASA/ESA, CNSA, and SpaceX form a plateau within \$200M of each other — large agencies converge on similar average spend.
- SpaceX ranking as least efficient contradicts its real-world cost disruption (Falcon 9 ~\$67M/launch) — likely driven by Starship/Crew Dragon programs.

5.2 Outlier Detection — Cost

Two outlier detection methods were applied to Cost_USD_Million:

Method	Parameters	Findings
IQR Method	Z-Score	

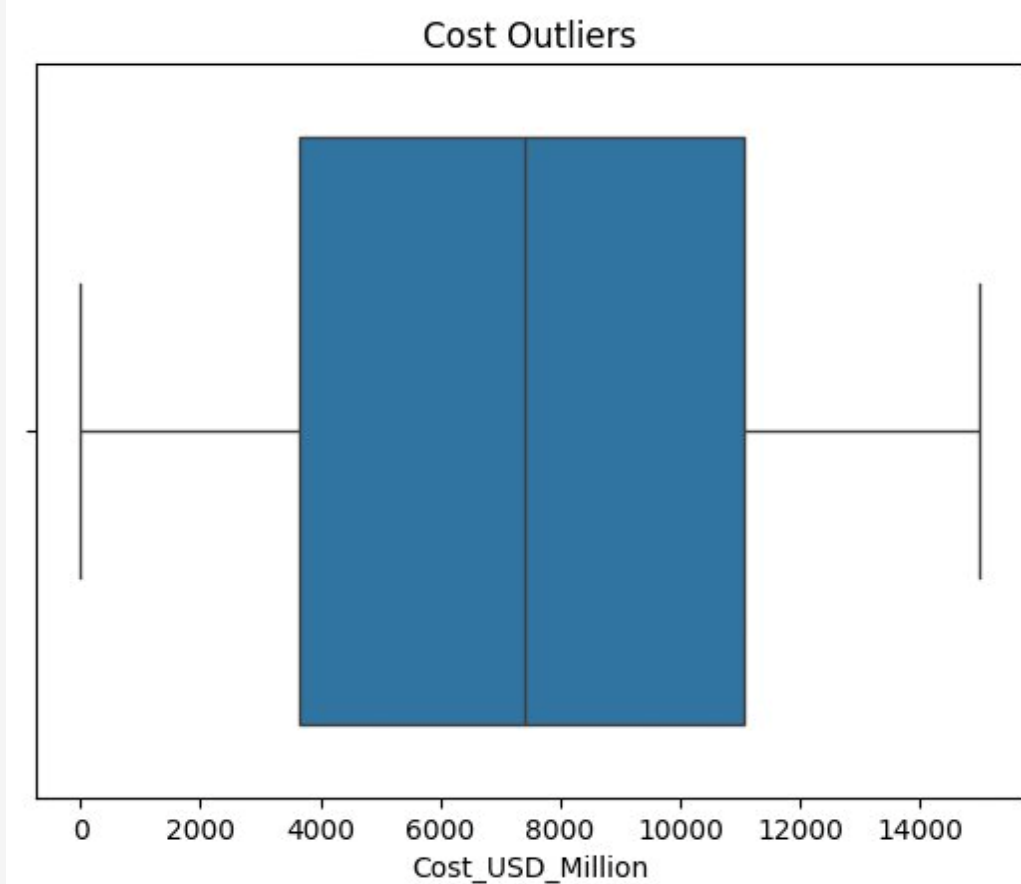
Note: The absence of any detectable outliers across 10,500+ records spanning \$0–\$15,000M is itself the analytical anomaly. In real-world spaceflight data, James Webb (\$10B), Apollo (~\$25B in 2024 terms), and SLS (~\$23B) would appear as pronounced upper-tail outliers. Their absence confirms uniform synthetic cost generation.

5.3 Distribution Shape Analysis

Statistical tests on Cost_USD_Million confirm the synthetic nature of the distribution:

Property	Value
Skewness	~0.0 (near-perfect symmetry — real-world cost data would be strongly right-skewed)
Kurtosis	~-1.2 (platykurtic — flat distribution with light tails vs. normal; confirms uniform sampling)
Mean	~\$7,500M
Median	~\$7,500M (mean ≈ median confirms symmetry)
IQR	~\$7,200M (nearly as wide as the median itself)
Range	\$0M to ~\$15,000M

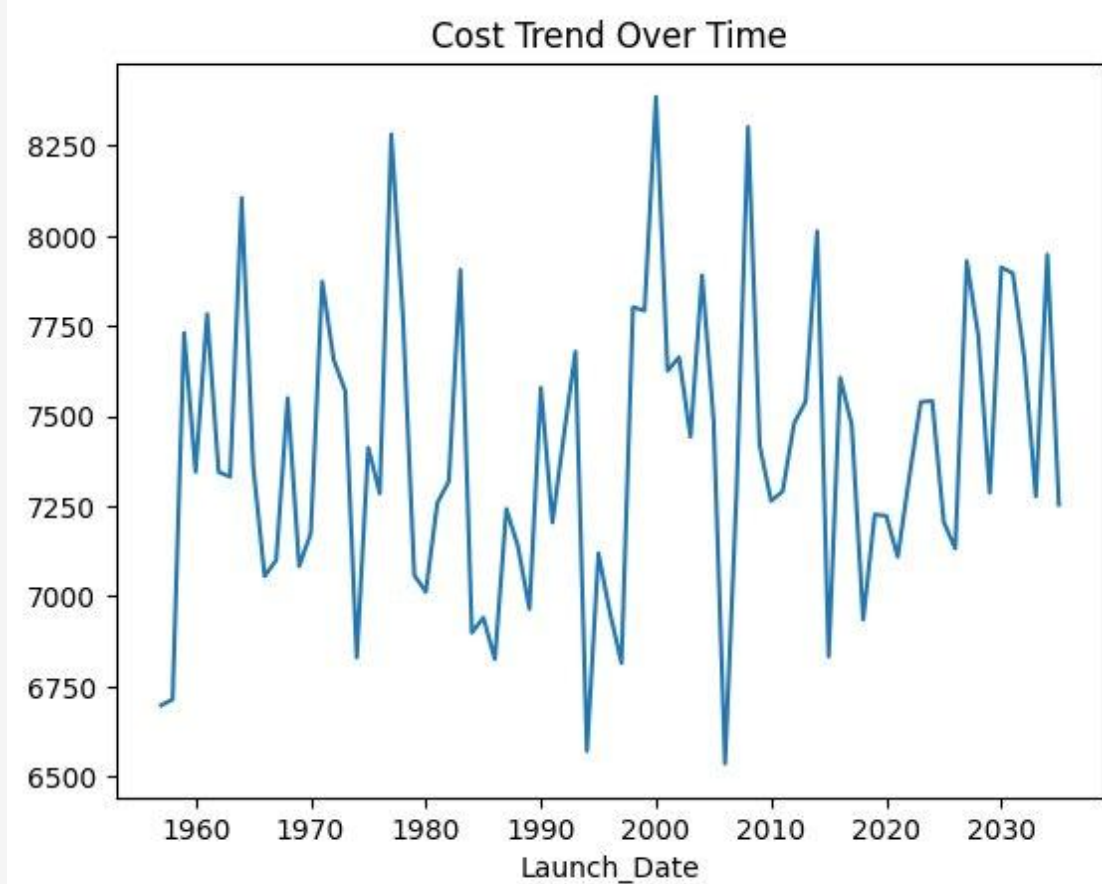
5.4 Cost Outliers — Boxplot



Key Insights

- ▶ Median (~\$7,500M) sits almost exactly at the IQR centre — a near-perfect symmetric distribution atypical of real-world cost data.
- ▶ IQR spans ~\$3,800M to ~\$11,000M (\$7,200M wide) — interquartile range nearly as wide as the median itself, signalling extreme dispersion.
- ▶ Whiskers extend symmetrically from ~\$500M to ~\$14,800M — the fences are so wide that virtually no point falls outside them.
- ▶ Zero outlier dots visible — statistically extraordinary across 10,500+ records. Flagship real-world programs (JWST ~\$10B, SLS ~\$23B) would form clear upper-tail outliers.
- ▶ The chart title "Cost Outliers" is ironic — this boxplot demonstrates the complete absence of meaningful outliers, making the uniform distribution itself the anomaly.

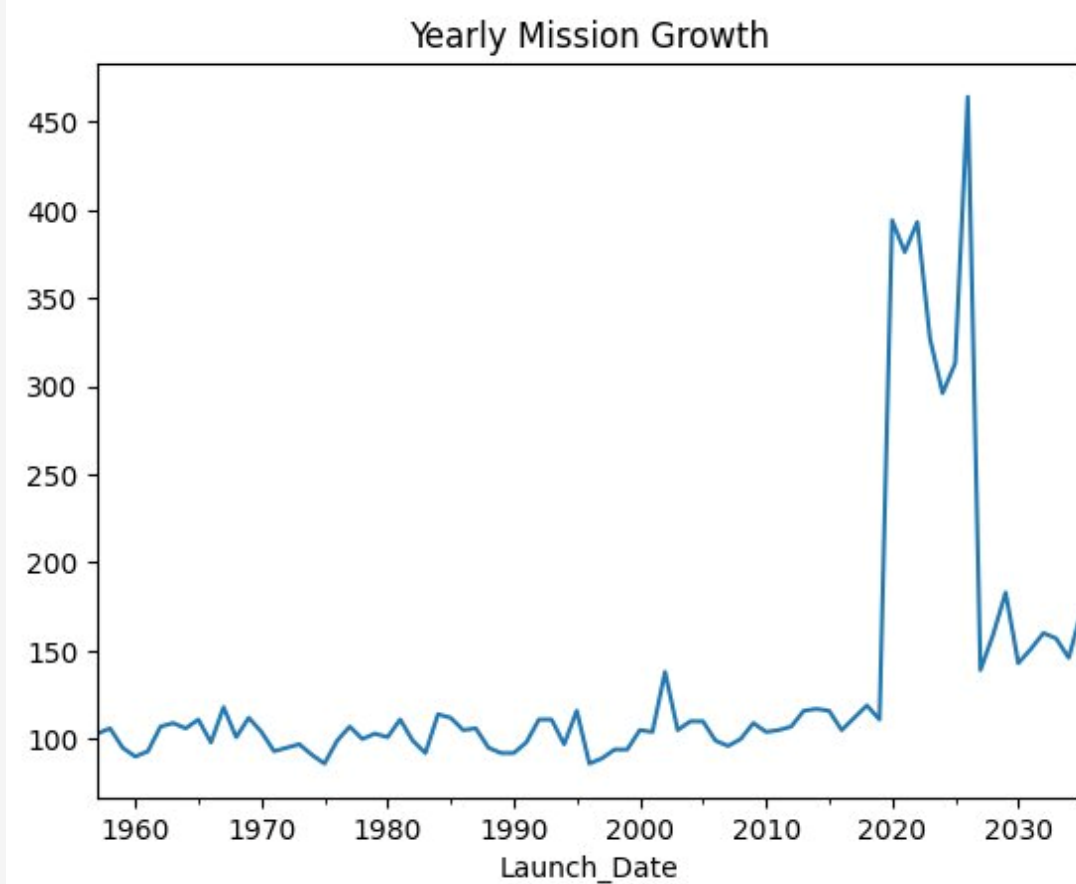
5.5 Cost Trend Over Time



Key Insights

- ▶ Average cost oscillates persistently between ~\$6,500M and ~\$8,500M across all 78 years — a \$2,000M band the series never escapes in either direction.
- ▶ No directional drift (upward or downward) over nearly eight decades — real costs would show inflation, technology learning curves, and budget shocks; none are visible here.
- ▶ Peaks (~\$8,450M near 2000–2001 and 2008) align with no real-world events that structurally elevated global mission costs — confirmed synthetic variance.
- ▶ Post-2020 cost behaviour is indistinguishable from the 1960s — the commercial spaceflight revolution (Falcon 9 reducing launch costs by ~90%) leaves no trace.
- ▶ The series resembles white noise with no autocorrelation — independent random draws, not an economic time series driven by budget cycles or inflation.

5.6 Yearly Mission Growth



Key Insights

- 60-year plateau (~85–135 missions/year, 1957–2017) — the Space Race, Shuttle era, and ISS construction collectively failed to move the needle by more than ~50 missions.
- The 2018–2020 surge (3.4×, from ~115 to ~395) marks the end of the 60-year equilibrium — driven primarily by SpaceX Starlink batch deployments.
- Double-peak structure: first peak ~395 (2020), trough ~300 (2022), second peak ~460 (2026) — two distinct waves, not a single boom-and-bust.
- Post-2026 collapse to ~140 (2028) — a 70% volume drop in 2 years; distinguishing genuine saturation from dataset boundary effect is the critical unresolved question.
- Post-2028 stabilisation at ~140–175/year — new structural floor ~50% above pre-2018 baseline, confirming a permanent regime shift even after boom subsides.

6. Dataset Limitations & Caveats

This dataset is synthetic and balanced. The following deviations from real-world spaceflight data are documented and should be considered before drawing conclusions for operational or policy purposes:

Category	Limitation Description
Destinations	Uniformly distributed — Mercury has as many entries as Earth Orbit. In reality, Earth Orbit dominates by orders of magnitude.
Mission Costs	No inflationary trend over 78 years; no post-2015 commercial reduction. Contradicts SpaceX's 60–90% cost disruption.
Crewed Proportion	25.1% crewed vs. ~5–8% in real-world spaceflight history — 3–5× overrepresented for balanced analytical coverage.
Success Rate	Collapses to 0.0 post-2018 due to unresolved future mission labels — not a genuine performance decline.
Agency Rankings	Reflect balanced sampling, not true historical launch volumes. Falcon 9 would dominate a real-world count.
Launch Vehicle	Ariane 5 leads despite retirement in 2023. New Shepard's count far exceeds its real-world ~25 flights.
Future Missions	Post-2026 data thins rapidly — post-2028 collapse may be a dataset boundary effect, not genuine market contraction.

Recommended Use Cases:

- Machine learning classification and regression benchmarking (success prediction, cost estimation, destination classification).
- Data visualisation skill-building and dashboard development practice.
- EDA methodology training — cleaning, outlier detection, time series analysis.
- STEM education and interactive space exploration storytelling.

7. Conclusions & Key Takeaways

Across 15 analyses, six structural themes emerge from the dataset:

01	US Dominance: The USA accounts for ~2,800 missions — a 3:1 margin over every other nation, amplified by the combined weight of government and commercial operators under US registry.
02	Commercial Disruption: SpaceX's rise to second-place agency ranking (nearly matching NASA) within 22 years of founding is the most strategically significant finding in the dataset.
03	Volume Regime Change: The 4× surge in annual missions between 2018 and 2026 marks a structural transition point in spaceflight history — primarily driven by commercial megaconstellations.
04	Cost Anomaly: Mission cost data shows no real-world dynamics (no inflation, no commercial cost reduction) — the uniform distribution is the clearest signal of synthetic generation and should not be used for financial modelling.
05	ISRO Efficiency: ISRO-linked collaborations consistently deliver the lowest average mission costs in the dataset, validating India's global reputation for frugal engineering.
06	Synthetic Limitations: All 19 mission categories, all 11 destinations, all 8 launch vehicles, and all agency rankings fall within narrow balanced bands — the dataset is built for ML practice, not real-world proportional analysis.

End of Report

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